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MIXTURES AND ALLIGATIONS

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Question

In a mixture of 56 litres, the ratio of acid to base is 5 : 2. How much base must be added to make the mixture ratio 5 :3?

A) 10 lts

B) 4 lts

C) 8 lts

D) 15 lts

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Question

The ratio of sodium chloride to ammonium in 100 kg of mixed ammonium normally used by three chemists is 7: 27. The amount of sodium chloride to be added to 100 kg of mixed ammonium to make the ratio 10 :27 is _____.

A) 6.7 kg

B) 8.8 kg

C) 9 kg

D) 11.1 kg

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Question 6

An alloy contains only zinc and copper. One such alloy weighing 21-gm contains zinc and copper in the ratio of 2: 5 by weight. If 30 gm of zinc is added then find what amount of copper has to be removed from the alloy such that the final alloy has zinc and copper in the ratio of 6:1 by weight.

A) 5 gm

B) 9 gm

C) 15 gm

D) 18 gm

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Question 3

Mr X mixed 15 kg of variety A rice with 20 kg of variety B rice and sold the mixture at a price 40% more than that of A. He did not get any profit. What is the ratio of the cost price of variety A to that of B per kg?

A) 10:17

B) 12:13

C) 17:10

D) 13:25

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Question 7

Three bottles contain equal mixtures of spirit and water in the ratio 4 : 1, 3 : 2 and 2 : 1 respectively. If all the solutions are mixed together, the ratio of spirit to water in the final mixture will be _____.

A) 15:17

B) 23:27

C) 31:14

D) 12:15

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Question

How much sugar must be mixed in 250 ml of sugar solution which contains 30% sugar by Weight, such that it becomes a 50% sugar solution?

A) 50 ml

B) 75 ml

C) 90 ml

D) 100 ml

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Question 8

A vessel contains a 120 ml mixture of milk and water. 2/5 of the mixture is milk, the remainder is water. An additional quantity of 168 ml of milk is poured into the mixture. What is the ratio of the volume of milk to water in the new mixture?

A) 1:4

B) 1:3

C) 3:1

D) 4:1

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Question 9

Two jars contain milk and water in the ratio 7: 4 and 3: 1 respectively. What volume should be taken out from the first jar if volumes have to be taken out from both jars so as to fill up a third 30 lit jar with milk to water in the ratio 1: 1 ?

A) 7.5 lit

B) 9.25 lit

C) 12 lit

D) It is impossible

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Question 2

Ajay pays Rs 14.4 per litre of spirit, He adds water and sells the mixture at Rs 12 per Iitre, thereby making 22.5% profit. Find the proportion of the water to that of the spirit received by the customers.

A) 27:35

B) 47:100

C) 100:47

D) 12:27

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Question 10

One type of liquid contains 24% of benzene, the other contains 36% of benzene. A can is filled with 5 parts of the first liquid and 7 parts of the second liquid. Find the percentage of benzene in the new mixture.

A) 30%

B) 31%

C) 34%

D) 35%

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Question 11

Nitrous oxide (laughing gas) worth Rs 136 per kg and Rs 145 per kg are mixed with a third variety in the ratio 1 : 1 : 3. If the mixture is worth Rs 163 per kg, then the price of the third variety per kg will be _____.

A) Rs. 178

B) Rs. 178.5

C) Rs. 180.5

D) Rs. 185

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Question 12

Two equal containers are filled with a mixture of water and petrol. One of them contains three times as much petrol as in the other. The mixtures in the two containers are then mixed and it is found that the ratio of water to petrol is 3: 2. Find the ratio of water to petrol in each of the original containers.

A) 2:1, 3:4

B) 1:3, 1:2

C) 2: 3, 4: 1

D) None of the above

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Question 13

A milkman had 20 litres of pure milk. He removed some milk and added an equal amount of water. He repeated the process again. Now, the milk was only 64% pure. Find the amount of milk that he removed initially.

A) 4 litres

B) 6 litres

C) 6.6 litres

D) 5 litres

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Question 14

Three beakers namely A, B and C each contain 100 ml of spirit in water solution. The ratio of spirit to water in the beakers A, B and C is 1 : 3, 1 : 4 and 2 : 3 respectively. 40 ml of solution is transferred from beaker A to beaker C and than 28 ml of solution is transferred from beaker C to beaker B. Find the final ratio of spirit in the beakers A, B and C?

A) 3 : 6 : 8

B) 6 : 15 : 20

C) 15 : 28 : 42

D) None of these

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Question 15

In a research experiment, a scientist is trying to invent a new compound which will have 30% concentration of nitrogen by mixing two mixtures of nitrogen, hydrogen and oxygen in the ratio 5 : 3. The ratio of these gases in the first mixture is 4 : 5 : 1 and in the second mixture is 5 : 7 : 3. Having found that the resultant concentration of nitrogen in the mixture is not 30%, he dilutes it with more oxygen to make it 30%. Find the ratio of the original
mixture to the oxygen which was used for dilution.

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Question 15

A) 4:2

B) 4:4

C) 4:1

D) 4:7

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THANK YOU

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